

VELORA — Database Architecture Part 1: Core, Auth, Subscription, POS

All PKs are UUID v7. All tenant-scoped tables have `business_id`. Money = `BIGINT` (smallest unit).

Global Conventions

Convention	Rule
Primary Key	<code>id</code> UUID v7, <code>CHAR(36)</code>
Foreign Key	<code>{table_singular}_id</code> <code>CHAR(36)</code>
Timestamps	<code>created_at</code> , <code>updated_at</code> (<code>TIMESTAMP</code>)
Soft Delete	<code>deleted_at</code> <code>TIMESTAMP</code> NULL where needed
Money	<code>BIGINT UNSIGNED</code> — store in smallest unit (IDR = no decimal, store as integer)
Boolean	<code>TINYINT(1)</code> with <code>is_</code> or <code>has_</code> prefix
Enum	Laravel PHP Enum backed by <code>VARCHAR</code>
Index	Every FK indexed. Composite index on <code>(business_id, ...)</code> for tenant queries

Module: Tenant (Core SaaS)

users

Column	Type	Constraints	Description
<code>id</code>	<code>CHAR(36)</code>	PK	UUID v7
<code>name</code>	<code>VARCHAR(100)</code>	NOT NULL	Full name

Column	Type	Constraints	Description
email	VARCHAR(150)	UNIQUE, NOT NULL	Login email
phone	VARCHAR(20)	NULLABLE, INDEX	Phone number
password	VARCHAR(255)	NOT NULL	Hashed
avatar_url	VARCHAR(500)	NULLABLE	Profile picture
email_verified_at	TIMESTAMP	NULLABLE	Email verification
is_platform_admin	TINYINT(1)	DEFAULT 0	Super admin flag
last_login_at	TIMESTAMP	NULLABLE	Last login tracking
last_login_ip	VARCHAR(45)	NULLABLE	IPv4/IPv6
status	VARCHAR(20)	DEFAULT 'active'	Enum: active, suspended, banned
created_at	TIMESTAMP		
updated_at	TIMESTAMP		
deleted_at	TIMESTAMP	NULLABLE	Soft delete

businesses

Column	Type	Constraints	Description
id	CHAR(36)	PK	
owner_id	CHAR(36)	FK→users, NOT NULL	Business owner
name	VARCHAR(150)	NOT NULL	Business name
slug	VARCHAR(150)	UNIQUE, NOT NULL	URL-friendly identifier
legal_name	VARCHAR(200)	NULLABLE	Legal entity name
tax_id	VARCHAR(50)	NULLABLE	NPWP

Column	Type	Constraints	Description
business_type	VARCHAR(30)	NOT NULL	Enum: retail, fnb, service, wholesale
industry	VARCHAR(50)	NULLABLE	e.g., "Food & Beverage"
phone	VARCHAR(20)	NULLABLE	
email	VARCHAR(150)	NULLABLE	
website	VARCHAR(255)	NULLABLE	
logo_url	VARCHAR(500)	NULLABLE	
address	TEXT	NULLABLE	
city	VARCHAR(100)	NULLABLE	
province	VARCHAR(100)	NULLABLE	
postal_code	VARCHAR(10)	NULLABLE	
country	VARCHAR(5)	DEFAULT 'ID'	ISO country code
currency	VARCHAR(5)	DEFAULT 'IDR'	Primary currency
timezone	VARCHAR(50)	DEFAULT 'Asia/Jakarta'	
status	VARCHAR(20)	DEFAULT 'active'	Enum: active, suspended, locked
locked_at	TIMESTAMP	NULLABLE	When subscription locks
locked_reason	VARCHAR(100)	NULLABLE	
settings	JSON	NULLABLE	Business-level settings
created_at	TIMESTAMP		
updated_at	TIMESTAMP		
deleted_at	TIMESTAMP	NULLABLE	

Indexes: (owner_id) , (slug) , (status)

branches

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses, NOT NULL	
name	VARCHAR(100)	NOT NULL	Branch name
code	VARCHAR(20)	NOT NULL	Short code e.g., "HQ", "CB-01"
type	VARCHAR(20)	DEFAULT 'store'	Enum: store, warehouse, hybrid
phone	VARCHAR(20)	NULLABLE	
email	VARCHAR(150)	NULLABLE	
address	TEXT	NULLABLE	
city	VARCHAR(100)	NULLABLE	
province	VARCHAR(100)	NULLABLE	
postal_code	VARCHAR(10)	NULLABLE	
latitude	DECIMAL(10,7)	NULLABLE	
longitude	DECIMAL(10,7)	NULLABLE	
is_main	TINYINT(1)	DEFAULT 0	Main/HQ branch
is_active	TINYINT(1)	DEFAULT 1	
settings	JSON	NULLABLE	Branch-level settings
created_at	TIMESTAMP		
updated_at	TIMESTAMP		
deleted_at	TIMESTAMP	NULLABLE	

Indexes: (business_id, is_active) , UNIQUE(business_id, code)

business_user (Pivot)

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
user_id	CHAR(36)	FK→users	
role_id	CHAR(36)	FK→roles	Primary role
is_owner	TINYINT(1)	DEFAULT 0	
joined_at	TIMESTAMP	NOT NULL	
status	VARCHAR(20)	DEFAULT 'active'	Enum: active, invited, suspended
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: UNIQUE(business_id, user_id)

branch_user (Pivot)

Column	Type	Constraints	Description
id	CHAR(36)	PK	
branch_id	CHAR(36)	FK→branches	
user_id	CHAR(36)	FK→users	
is_default	TINYINT(1)	DEFAULT 0	Default branch for user
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: UNIQUE(branch_id, user_id)

Module: Subscription

subscription_plans

Column	Type	Constraints	Description
id	CHAR(36)	PK	
name	VARCHAR(100)	NOT NULL	e.g., "Starter", "Pro", "Enterprise"
slug	VARCHAR(100)	UNIQUE, NOT NULL	
description	TEXT	NULLABLE	
price_monthly	BIGINT UNSIGNED	DEFAULT 0	Monthly price
price_quarterly	BIGINT UNSIGNED	DEFAULT 0	3-month price
price_biannual	BIGINT UNSIGNED	DEFAULT 0	6-month price
price_annual	BIGINT UNSIGNED	DEFAULT 0	Yearly price
trial_days	INT UNSIGNED	DEFAULT 0	Free trial period
grace_period_days	INT UNSIGNED	DEFAULT 3	Days after expiry before lock
is_active	TINYINT(1)	DEFAULT 1	
is_featured	TINYINT(1)	DEFAULT 0	Highlighted plan
sort_order	INT	DEFAULT 0	Display order
metadata	JSON	NULLABLE	Extra plan data
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

feature_limits

Column	Type	Constraints	Description
id	CHAR(36)	PK	
plan_id	CHAR(36)	FK→subscription_plans	
feature_key	VARCHAR(50)	NOT NULL	e.g., "max_products", "max_branches", "max_users"
feature_value	VARCHAR(50)	NOT NULL	Number or "unlimited"
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: UNIQUE(plan_id, feature_key)

Feature keys include: max_products , max_branches , max_users , max_warehouses ,
max_transactions_per_month , enable_multi_warehouse , enable_batch_tracking , enable_loyalty ,
enable_api_access , enable_advanced_reports

subscriptions

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses, NOT NULL	
plan_id	CHAR(36)	FK→subscription_plans	
billing_cycle	VARCHAR(20)	NOT NULL	Enum: monthly, quarterly, biannual, annual
status	VARCHAR(20)	NOT NULL	Enum: trial, active, past_due, grace, expired, cancelled
trial_starts_at	TIMESTAMP	NULLABLE	
trial_ends_at	TIMESTAMP	NULLABLE	
starts_at	TIMESTAMP	NOT NULL	

Column	Type	Constraints	Description
ends_at	TIMESTAMP	NOT NULL	
grace_ends_at	TIMESTAMP	NULLABLE	Grace period end
cancelled_at	TIMESTAMP	NULLABLE	
cancel_reason	TEXT	NULLABLE	
auto_renew	TINYINT(1)	DEFAULT 1	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: (business_id, status) , (ends_at)

subscription_payments

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
subscription_id	CHAR(36)	FK→subscriptions	
invoice_number	VARCHAR(50)	UNIQUE, NOT NULL	e.g., "INV-2026-00001"
amount	BIGINT UNSIGNED	NOT NULL	Total amount
discount_amount	BIGINT UNSIGNED	DEFAULT 0	
tax_amount	BIGINT UNSIGNED	DEFAULT 0	
total_amount	BIGINT UNSIGNED	NOT NULL	Final amount
payment_method	VARCHAR(30)	NULLABLE	bank_transfer, ewallet, etc.

Column	Type	Constraints	Description
payment_gateway	VARCHAR(30)	NULLABLE	midtrans, xendit
gateway_transaction_id	VARCHAR(100)	NULLABLE	External ref
status	VARCHAR(20)	NOT NULL	Enum: pending, paid, failed, refunded, expired
paid_at	TIMESTAMP	NULLABLE	
due_at	TIMESTAMP	NOT NULL	
notes	TEXT	NULLABLE	
metadata	JSON	NULLABLE	Gateway response data
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

subscription_logs

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
subscription_id	CHAR(36)	FK→subscriptions	
action	VARCHAR(30)	NOT NULL	Enum: created, renewed, upgraded, downgraded, cancelled, expired, locked, unlocked
from_plan_id	CHAR(36)	NULLABLE	Previous plan
to_plan_id	CHAR(36)	NULLABLE	New plan
note	TEXT	NULLABLE	
performed_by	CHAR(36)	NULLABLE	FK→users
created_at	TIMESTAMP		

Module: Auth / Security

roles

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	NULLABLE	NULL = system role, else tenant role
name	VARCHAR(50)	NOT NULL	e.g., "owner", "admin", "cashier"
slug	VARCHAR(50)	NOT NULL	
display_name	VARCHAR(100)	NULLABLE	
description	VARCHAR(255)	NULLABLE	
is_system	TINYINT(1)	DEFAULT 0	System-defined, cannot delete
level	INT UNSIGNED	DEFAULT 0	Hierarchy: 0=highest
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: UNIQUE(business_id, slug)

System roles: platform_admin , owner , admin , manager , cashier , warehouse_staff

permissions

Column	Type	Constraints	Description
id	CHAR(36)	PK	
module	VARCHAR(30)	NOT NULL	e.g., "pos", "inventory", "finance"
name	VARCHAR(80)	UNIQUE, NOT NULL	e.g., "products.create"
display_name	VARCHAR(150)	NULLABLE	
description	VARCHAR(255)	NULLABLE	
created_at	TIMESTAMP		

Column	Type	Constraints	Description
updated_at	TIMESTAMP		

role_permission (Pivot)

Column	Type	Constraints	Description
role_id	CHAR(36)	FK→roles	
permission_id	CHAR(36)	FK→permissions	

Index: PRIMARY(role_id, permission_id)

activity_logs

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	NULLABLE, INDEX	
user_id	CHAR(36)	NULLABLE, FK→users	
action	VARCHAR(50)	NOT NULL	login, logout, create, update, delete
module	VARCHAR(30)	NULLABLE	Which module
description	VARCHAR(500)	NULLABLE	Human-readable
ip_address	VARCHAR(45)	NULLABLE	
user_agent	VARCHAR(500)	NULLABLE	
created_at	TIMESTAMP		

Indexes: (business_id, created_at) , (user_id, created_at)

audit_logs

Column	Type	Constraints	Description
id	CHAR(36)	PK	

Column	Type	Constraints	Description
business_id	CHAR(36)	INDEX	
user_id	CHAR(36)	NULLABLE	
auditable_type	VARCHAR(150)	NOT NULL	Model class
auditable_id	CHAR(36)	NOT NULL	Model ID
event	VARCHAR(20)	NOT NULL	created, updated, deleted, restored
old_values	JSON	NULLABLE	Before state
new_values	JSON	NULLABLE	After state
url	VARCHAR(500)	NULLABLE	
ip_address	VARCHAR(45)	NULLABLE	
created_at	TIMESTAMP		

Indexes: (auditable_type, auditable_id) , (business_id, created_at)

Module: POS (Products)

categories

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
parent_id	CHAR(36)	NULLABLE, FK→categories	Nested categories
name	VARCHAR(100)	NOT NULL	
slug	VARCHAR(100)	NOT NULL	
description	TEXT	NULLABLE	
image_url	VARCHAR(500)	NULLABLE	

Column	Type	Constraints	Description
sort_order	INT	DEFAULT 0	
is_active	TINYINT(1)	DEFAULT 1	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		
deleted_at	TIMESTAMP	NULLABLE	

Indexes: (business_id, is_active) , UNIQUE(business_id, slug) , (parent_id)

brands

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
name	VARCHAR(100)	NOT NULL	
slug	VARCHAR(100)	NOT NULL	
logo_url	VARCHAR(500)	NULLABLE	
is_active	TINYINT(1)	DEFAULT 1	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		
deleted_at	TIMESTAMP	NULLABLE	

Indexes: UNIQUE(business_id, slug)

units

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	NULLABLE	NULL = system unit

Column	Type	Constraints	Description
name	VARCHAR(50)	NOT NULL	e.g., "Kilogram"
abbreviation	VARCHAR(10)	NOT NULL	e.g., "kg"
group	VARCHAR(30)	NOT NULL	Enum: weight, volume, length, quantity, custom
is_system	TINYINT(1)	DEFAULT 0	System-defined
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

System units seeded: pcs, pack, box, dus, karton, lusin, kg, gram, mg, liter, ml, meter, cm, botol, galon, lembar, batang

unit_conversions

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
from_unit_id	CHAR(36)	FK→units	
to_unit_id	CHAR(36)	FK→units	
conversion_factor	DECIMAL(15,6)	NOT NULL	1 from_unit = X to_unit
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: UNIQUE(business_id, from_unit_id, to_unit_id)

Example: from=dus, to=pcs, factor=24 → 1 dus = 24 pcs

products

Column	Type	Constraints	Description
id	CHAR(36)	PK	

Column	Type	Constraints	Description
business_id	CHAR(36)	FK→businesses	
category_id	CHAR(36)	NULLABLE, FK→categories	
brand_id	CHAR(36)	NULLABLE, FK→brands	
name	VARCHAR(200)	NOT NULL	
slug	VARCHAR(200)	NOT NULL	
sku	VARCHAR(50)	NULLABLE	Stock Keeping Unit
barcode	VARCHAR(50)	NULLABLE	Primary barcode
description	TEXT	NULLABLE	
base_unit_id	CHAR(36)	FK→units	Smallest unit for this product
purchase_price	BIGINT UNSIGNED	DEFAULT 0	Default buy price (in base unit)
selling_price	BIGINT UNSIGNED	DEFAULT 0	Default sell price (in base unit)
minimum_price	BIGINT UNSIGNED	DEFAULT 0	Floor price (for discounts)
product_type	VARCHAR(20)	DEFAULT 'standard'	Enum: standard, variant, service, bundle
tax_type	VARCHAR(20)	DEFAULT 'taxable'	Enum: taxable, non_taxable, inclusive
is_active	TINYINT(1)	DEFAULT 1	
is_track_stock	TINYINT(1)	DEFAULT 1	Enable stock tracking
is_track_batch	TINYINT(1)	DEFAULT 0	Enable batch/lot tracking
is_track_expiry	TINYINT(1)	DEFAULT 0	Enable expiry tracking
has_variants	TINYINT(1)	DEFAULT 0	Has size/color variants

Column	Type	Constraints	Description
min_stock	INT	DEFAULT 0	Minimum stock alert
max_stock	INT	NULLABLE	Maximum stock
weight	DECIMAL(10,3)	NULLABLE	For shipping (grams)
image_url	VARCHAR(500)	NULLABLE	Primary image
images	JSON	NULLABLE	Array of image URLs
sort_order	INT	DEFAULT 0	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		
deleted_at	TIMESTAMP	NULLABLE	

Indexes: (business_id, is_active) , (business_id, category_id) , (business_id, sku) , (business_id, barcode) , FULLTEXT(name, sku, barcode)

product_variants

Column	Type	Constraints	Description
id	CHAR(36)	PK	
product_id	CHAR(36)	FK→products	
business_id	CHAR(36)	FK→businesses	
name	VARCHAR(100)	NOT NULL	e.g., "Large - Red"
sku	VARCHAR(50)	NULLABLE	Variant SKU
barcode	VARCHAR(50)	NULLABLE	
purchase_price	BIGINT UNSIGNED	DEFAULT 0	Override parent
selling_price	BIGINT UNSIGNED	DEFAULT 0	
min_stock	INT	DEFAULT 0	
image_url	VARCHAR(500)	NULLABLE	

Column	Type	Constraints	Description
option_values	JSON	NOT NULL	{"size":"Large","color":"Red"}
is_active	TINYINT(1)	DEFAULT 1	
sort_order	INT	DEFAULT 0	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		
deleted_at	TIMESTAMP	NULLABLE	

Indexes: (product_id) , (business_id, sku) , (business_id, barcode)

product_units (Multi-unit pricing per product)

Column	Type	Constraints	Description
id	CHAR(36)	PK	
product_id	CHAR(36)	FK→products	
unit_id	CHAR(36)	FK→units	
conversion_to_base	DECIMAL(15,6)	NOT NULL	1 this unit = X base units
purchase_price	BIGINT UNSIGNED	DEFAULT 0	Buy price in this unit
selling_price	BIGINT UNSIGNED	DEFAULT 0	Sell price in this unit
barcode	VARCHAR(50)	NULLABLE	Barcode for this unit
is_purchase_unit	TINYINT(1)	DEFAULT 0	Used for purchasing
is_sale_unit	TINYINT(1)	DEFAULT 1	Used for selling
is_default	TINYINT(1)	DEFAULT 0	Default display unit
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: UNIQUE(product_id, unit_id)

Example: Product "Aqua 600ml"

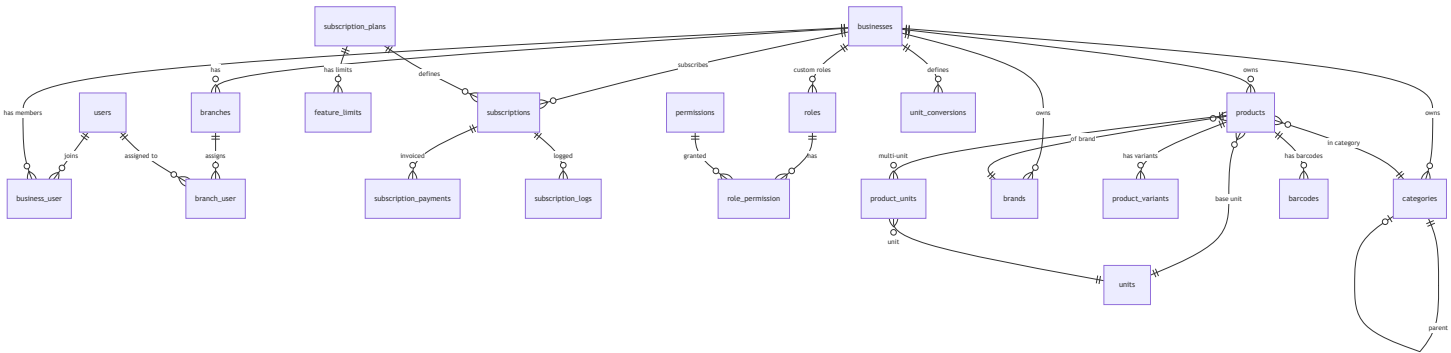
- base_unit = pcs
- product_units: {unit=dus, conversion=24, sell_price=60000}, {unit=karton, conversion=48, sell_price=115000}

barcodes (Additional barcodes per product)

Column	Type	Constraints	Description
id	CHAR(36)	PK	
business_id	CHAR(36)	FK→businesses	
barcodeable_type	VARCHAR(150)	NOT NULL	product or product_variant
barcodeable_id	CHAR(36)	NOT NULL	
code	VARCHAR(100)	NOT NULL	Barcode value
type	VARCHAR(20)	DEFAULT 'ean13'	Enum: ean13, ean8, upc, code128, qr, custom
unit_id	CHAR(36)	NULLABLE, FK→units	Barcode for specific unit
is_primary	TINYINT(1)	DEFAULT 0	
created_at	TIMESTAMP		
updated_at	TIMESTAMP		

Indexes: (business_id, code) , (barcodeable_type, barcodeable_id)

Relationship Diagram — Core + POS



Continued in [Part 2 — Inventory, Sales, Purchasing]

(file:///C:/Users/YP/.gemini/antigravity/brain/7880df49-c88e-4c8f-82bc-eb5a42cebb61/database_architecture_part2.md)

Continued in [Part 3 — CRM, Finance, Settings](file:///C:/Users/YP/.gemini/antigravity/brain/7880df49-c88e-4c8f-82bc-eb5a42cebb61/database_architecture_part3.md)